

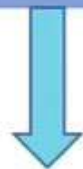
Memory Segmentation of 8086

- ❖ 8086 has 20 bits of physical address.
- ❖ Here, 20 bits of physical address is not byte compatible physical address.
- ❖ To avoid the issue of byte compatibility, Microprocessor 8086 uses memory segmentation. In which we use concept of virtual address.
- ❖ Here, memory is bisected into FOUR segments.
 - ☐ Code Segment
 - ☐ Stack Segment
 - ☐ Data Segment
 - ☐ Extra Segment
- ❖ To calculate physical address from virtual address we should know Segment address and OFFSET address.
- ❖ Segment Address is held by segment registers (16bits). [CS, SS, DS & ES]
- ❖ OFFSET Address is held by OFFSET pointers (16 bits). [IP, (SP & BP), SI & DI]
- ❖ So, Physical Address is calculated by
$$PA = SR \times 10H + OP$$
- ❖ As Offset Pointer is having size of 16bits, maximum size of any segment is $2^{16} = 2^6 \times 2^{10} = 64K$.
- ❖ Minimum size of any segment is $10H = 16\text{Bytes}$

Physical Address Calculation in 8086



Physical Address Calculation in 8086

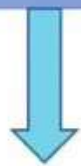


Physical Address of 20
bits $A_{19} - A_0$

- ❑ SP (Stack Pointer) – SP points top of stack segment.
- ❑ BP (Base Pointer) – BP points any address of stack segment.
- ❑ Physical Address of 8086 is given by

$$PA = SR \times 10H + OP$$

Physical Address Calculation in 8086



Physical Address of 20 bits $A_{19} - A_0$

- ☐ SP (Stack Pointer) – SP points top of stack segment.
- ☐ BP (Base Pointer) – BP points any address of stack segment.
- ☐ Physical Address of 8086 is given by

$$PA = SR \times 10H + OP$$

- ☐ Example 1 : If Code segment register is 1245H and Instruction Pointer is 1561H. Then finds the physical address of given instruction.

Physical Address Calculation in 8086



- ❑ SP (Stack Pointer) – SP points top of stack segment.
- ❑ BP (Base Pointer) – BP points any address of stack segment.
- ❑ Physical Address of 8086 is given by

$$PA = SR \times 10H + OP$$

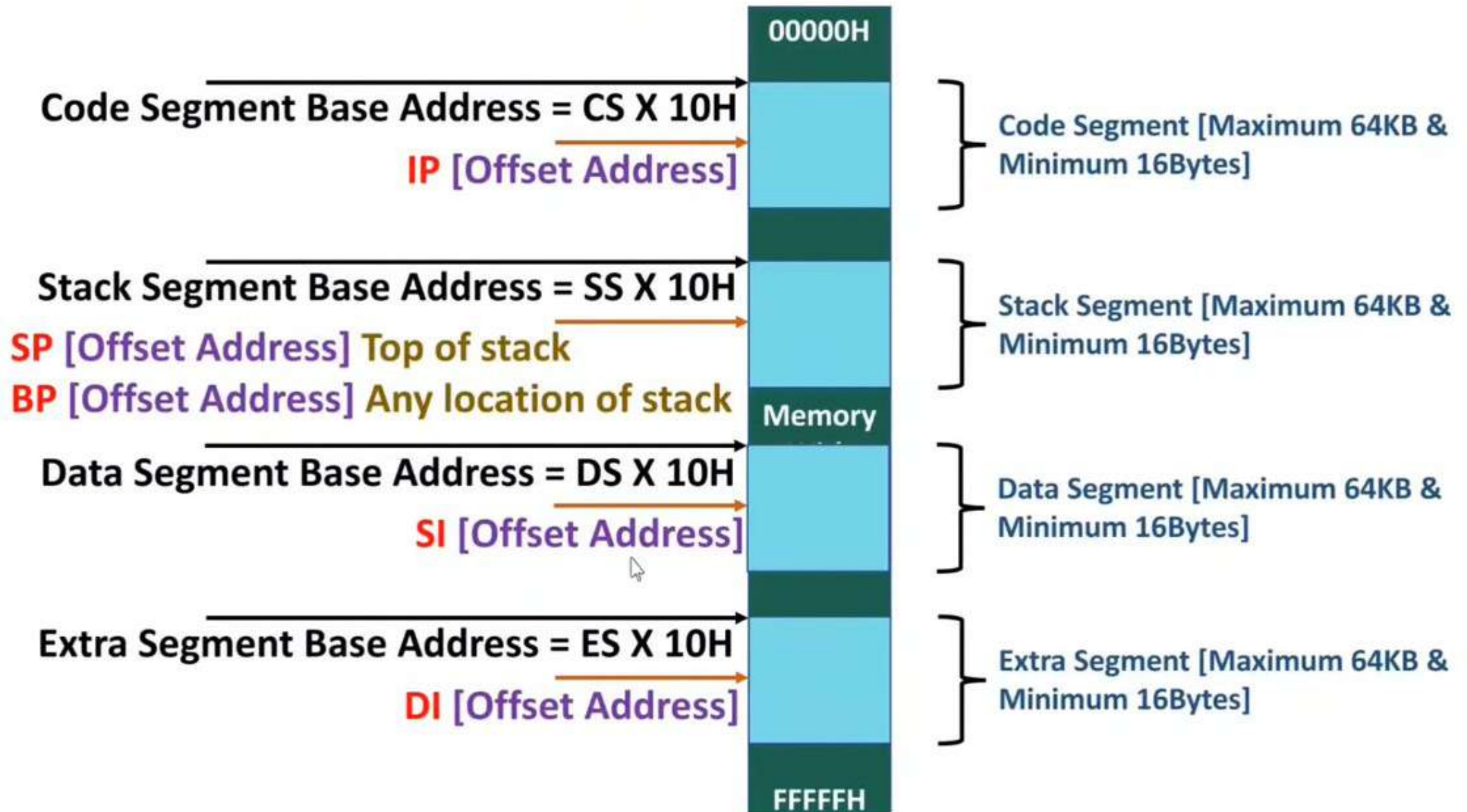
- ❑ Example 1 : If Code segment register is 1245H and Instruction Pointer is 1561H. Then finds the physical address of given instruction.

$$PA = SR \times 10H + OP$$

$$PA = 1245 \times 10H + 1561H = 12450H + 1561H$$

$$PA = 139B1H$$

Memory Segmentation of 8086



Physical Address Calculation in 8086



- ❑ SP (Stack Pointer) – SP points top of stack segment.
- ❑ BP (Base Pointer) – BP points any address of stack segment.
- ❑ Physical Address of 8086 is given by

$$PA = SR \times 10H + OP$$

- ❑ Example 1 : If Code segment register is 1245H and Instruction Pointer is 1561H. Then finds the physical address of given instruction.

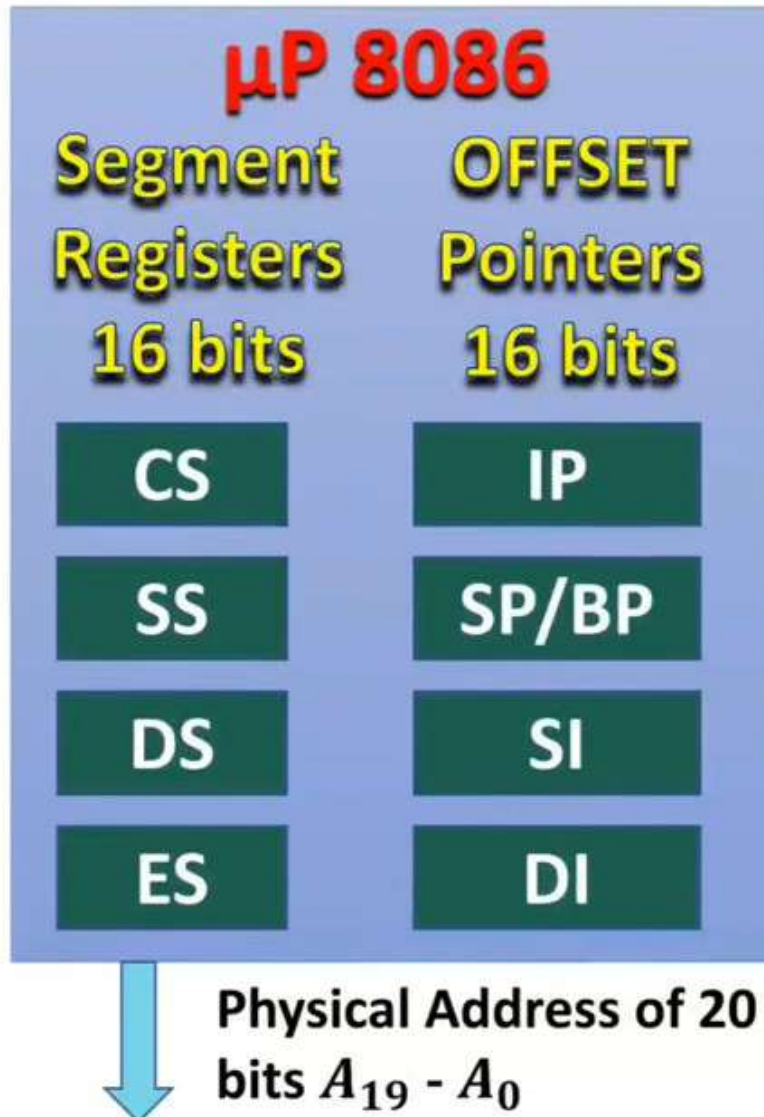
$$PA = SR \times 10H + OP$$

$$PA = 1245 \times 10H + 1561H = 12450H + 1561H$$

$$PA = 139B1H$$

- ❑ Example 2 : If Data segment register is 78EFH. Then finds the maximum and minimum range of physical address.

Physical Address Calculation in 8086



- ❑ SP (Stack Pointer) – SP points top of stack segment.
- ❑ BP (Base Pointer) – BP points any address of stack segment.
- ❑ Physical Address of 8086 is given by

$$PA = SR \times 10H + OP$$

- ❑ Example 1 : If Code segment register is 1245H and Instruction Pointer is 1561H. Then finds the physical address of given instruction.

$$PA = SR \times 10H + OP$$

$$PA = 1245 \times 10H + 1561H = 12450H + 1561H$$

$$PA = 139B1H$$

- ❑ Example 2 : If Data segment register is 78EFH. Then finds the maximum and minimum range of physical address.

- For Maximum range

$$PA = SR \times 10H + OP$$

$$PA = 78EFH \times 10H + (0000H \text{ to } FFFFH)$$

$$PA = 78EF0H \text{ to } 88EEFH$$

- For Minimum range

$$PA = SR \times 10H + OP$$

$$PA = 78EFH \times 10H + 0000H \text{ to } 000FH$$

$$PA = 78EF0H \text{ to } 78EFFH$$

Advantages of Memory Segmentation of 8086

- ❖ We can program 8086 with 16bits addressing.
- ❖ Due to Memory segmentation, 8086 becomes backward compatible with 8085.
- ❖ Change in One segment does not effect the other segment.
- ❖ Memory management gets easier.
- ❖ Debugging gets easier.
- ❖ It avoids memory overlap with different segments.

